

MACHINE LEARNING · BUSINESS INTELLIGENCE · PYTHON · STREAMLIT

Customer Intelligence Platform

Predict churn · Explain risk · Segment customers · Drive retention

Python

XGBoost

SHAP

Streamlit

DVC

scikit-learn

7,043

Customers

~0.85

ROC-AUC

26.54%

Churn Rate

THE PROBLEM

Customer churn is expensive — and nearly impossible to anticipate legacy style

26.54%

of customers churned in this telecom dataset

Revenue Drain

Every churned customer means lost recurring revenue and higher re-acquisition spend.

Reactive, Not Proactive

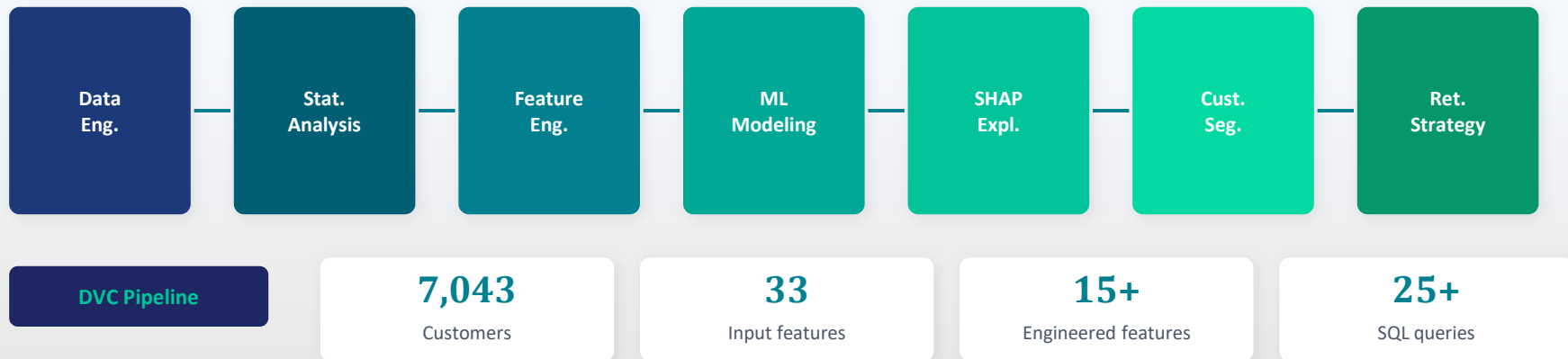
Traditional reports show churn after the fact — when it is already too late to intervene.

No Personalisation

Generic retention campaigns ignore each customer's unique risk profile and value.

END-TO-END PIPELINE

From raw data to business recommendations — 7 stages, 10 notebooks



Full pipeline version-controlled with DVC — reproducible, documented, modular Python src/ library architecture.

5 models benchmarked — best selected with F1-optimised thresholds

Logistic Regression

Baseline

Random Forest

Ensemble

XGBoost

Gradient Boost

LightGBM

Fast & Accurate

CatBoost

Top Performer

ROC-AUC Score

~0.85

Best-model cross-validated performance

F1-Optimized Threshold

Bypasses default 0.5 strict split

5-Fold Cross Validation

Ensures production generalization

SMOTE Resampling

Corrects severe class imbalance

Not just black-box predictions — explanations the business can trust



Global Insights

- ✓ Overall feature importance weights
- ✓ Macro variables pushing macro risk
- ✓ Enterprise risk element mapping
- ✓ Aggregated SHAP summary dependencies



Per-Customer Local Logic

- ✓ Why a specific user is flagged at risk
- ✓ Individual feature attribution forces
- ✓ Customized waterfall profiling blocks
- ✓ Direct pipeline stream into retention engines

5 distinct clusters × churn-adjusted CLV = clear priorities

Loyal Champions

High Value · Structural Churn Safetied

At-Risk Valuable

High Value · Impending High Churn Warning

Engaged Mid-Tier

Baseline Stable Value · Active Usage

Wavering Customers

Average Margin · Volatile Trend Metrics

Low-Engagement Churners

Low Acquisition Asset · High Loss State

Enhanced CLV Scoring Engine

Formula: Value = Projected CLV × (1 - Churn Prob.)

- Isolates High Value + High Risk targets instantly
- Establishes automated triage defense ranks
- Provides true financial context to model scores
- Calculates expected yield metrics per intervention

6-page production Streamlit app — live on the web

1 Portfolio Overview

Dataset distribution matrices, current baseline churn states, core KPI rollups.

2 Churn Prediction Engine

Real-time query processing for both individual vectors and bulk CSV batches.

3 SHAP Explanations

Interactive global importances and real-time local force attribution graphs.

4 Customer Segments

K-Means demographic analysis panels built directly via customer value profiles.

5 Lifetime Value Triage

Priority routing logic utilizing custom churn-adjusted risk scoring layers.

6 Business Intelligence

Over 25 complex analytical SQL query workflows mapping cohort retention charts.

App URL: [link](#)

OPEN SOURCE · FULLY DOCUMENTED · LIVE DEPLOYED

Explore the Platform

Machine learning that explains itself. Tactical recommendations a business can deploy today.
An end-to-end engineered asset — not just an isolated model.

[GitHub Code Base](#)

[Launch Live App](#)

Built with: Python · XGBoost · LightGBM · CatBoost · SHAP · scikit-learn · Streamlit · DVC · SQLite · Plotly

Engineered by Biswarup Majumdar · Open to New Opportunities